

# FoveaMap — Visual Design Document

Sep 30, 2026 · @Ammar Hashmi

## 1. Design goals

The dashboard must prove three claims within ten seconds of anyone looking at it: the map is detailed near the vehicle, it is small, and it is real time.

### Principles

1. **The map is the hero.** At least 60% of the screen is the live 2.5D map; every panel exists to explain it.
2. **Colour means class, nothing else.** One fixed palette for semantic classes across the map, legend, charts and the report. UI chrome stays neutral grey.
3. **Every number is live and sourced.** KPIs come from the same metrics stream as the benchmark report (Architecture section 10); nothing on screen is hard-coded.
4. **Show the fovea.** The tier boundary is always visible, so viewers see where resolution changes and that nothing breaks there.
5. **Compare, don't claim.** Memory and accuracy are always shown next to a baseline or target.
6. **Never slow the pipeline.** The dashboard drops its own frames before it ever delays the core (Architecture ADR-6).

## 2. Audiences and viewing contexts

Design for the projector first: if it reads on a 1080p projector from the back of a room, it reads everywhere.

| Context             | Viewer                         | Screen                           | Time spent  | What they must see                                                |
|---------------------|--------------------------------|----------------------------------|-------------|-------------------------------------------------------------------|
| Live demo / judging | Evaluator, audience            | 1920 × 1080 projector, dark room | 3–5 minutes | Fovea boundary, class colours, FPS, memory saving vs uniform      |
| Debugging session   | Perception or systems engineer | 14–16 in laptop, 1440p monitor   | Hours       | Layer toggles, cell inspector, per-stage latency, frame scrubbing |

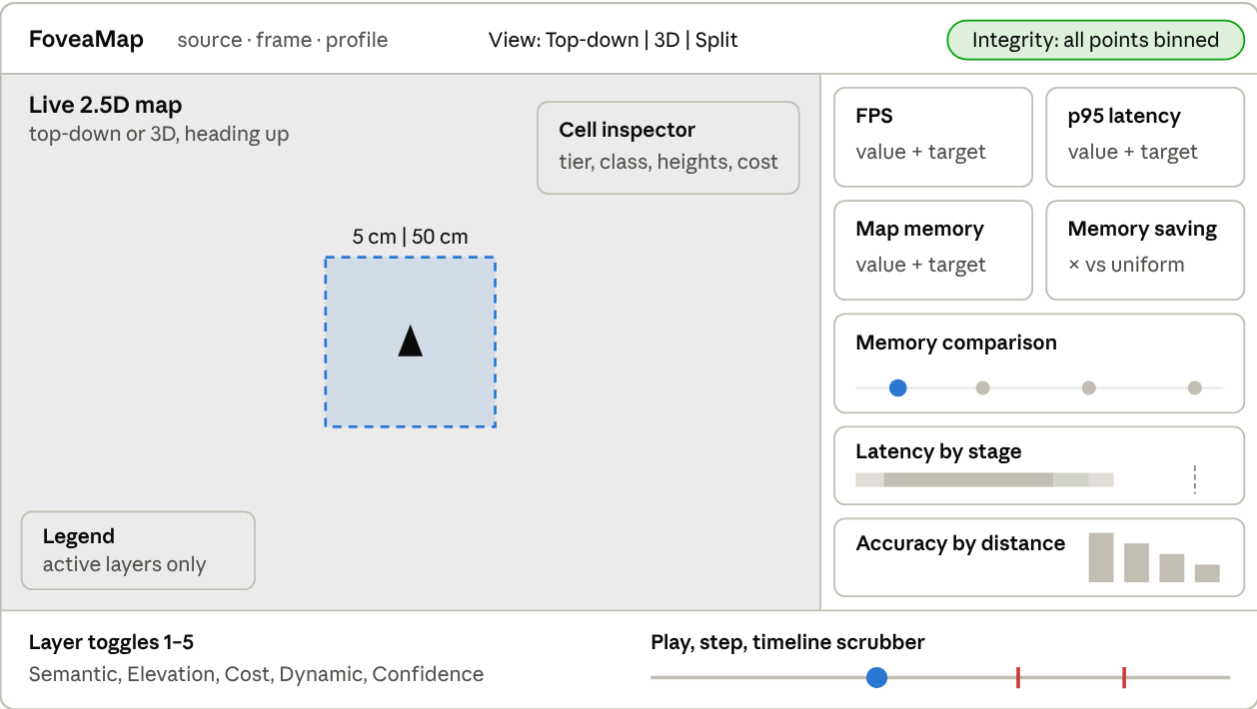
| Context        | Viewer             | Screen                   | Time spent    | What they must see                                     |
|----------------|--------------------|--------------------------|---------------|--------------------------------------------------------|
| Offline review | Researcher, mentor | Laptop or printed report | 10–30 minutes | Accuracy by distance band, failure frames, screenshots |

Default layout targets 1920 × 1080 in the dark theme; the light theme is for printed reports and daytime rooms.

### 3. Layout

One screen, no tabs: a top bar, the live map on the left two-thirds, a right rail of KPIs and charts, and a bottom bar for layers and playback.

The live map takes two-thirds of the screen; the right rail explains it



dashboard wireframe · 1920 × 1080, schematic

Red ticks on the timeline mark frames that failed a check (latency over target or points lost), so a presenter can jump straight to them.

| Region  | Size at 1920 × 1080 | Contents                                                               |
|---------|---------------------|------------------------------------------------------------------------|
| Top bar | full width × 56 px  | Name, data source + frame + grid profile, view switch, integrity badge |

| Region     | Size at 1920 × 1080 | Contents                                                                       |
|------------|---------------------|--------------------------------------------------------------------------------|
| Map        | 1250 × 900 px       | Live 2.5D map, fovea boundary, legend (bottom-left), cell inspector (floating) |
| Right rail | 640 × 900 px        | 4 KPI tiles, memory comparison, latency by stage, accuracy by distance         |
| Bottom bar | full width × 80 px  | Layer toggles, play / step, timeline scrubber                                  |

Below 1280 px wide the right rail moves under the map; the map never shrinks below 60% of the viewport height.

## 4. Semantic colour system

The palette is built on the Okabe-Ito colour-blind-safe set, and motion is shown by an outline and arrow rather than a new hue, so a class keeps its colour whether it moves or not.

| Map category        | Classes included                                     | Colour (hex)              | Fill on dark map             | Extra cue                              |
|---------------------|------------------------------------------------------|---------------------------|------------------------------|----------------------------------------|
| Drivable            | road, parking, lane marking                          | #009E73<br>bluish green   | 55% opacity                  | Hillshade shows curbs and potholes     |
| Non-drivable ground | sidewalk, terrain, other ground                      | #F0E442<br>yellow         | 30% opacity                  | —                                      |
| Static obstacle     | building, wall, fence, pole, sign, trunk, vegetation | #A7B1BD<br>cool grey      | 85%,<br>brighter with height | Height extrusion in 3D                 |
| Vehicle             | car, truck, bus, other vehicle                       | #E69F00<br>orange         | 100%                         | Box outline                            |
| Person / cyclist    | pedestrian, cyclist, motorcyclist                    | #CC79A7<br>reddish purple | 100%                         | Dot marker so small blobs stay visible |
| Moving (any agent)  | vehicle or person flagged moving                     | class colour              | 100%                         | 2 px white outline + velocity arrow    |

| Map category   | Classes included                | Colour (hex)     | Fill on dark map | Extra cue              |
|----------------|---------------------------------|------------------|------------------|------------------------|
| Unknown        | fewer than 3 points in 5 frames | #1A1F27          | 100%             | 45° hatch at 8% white  |
| Fovea boundary | tier edges                      | #56B4E9 sky blue | line only        | Dashed, labelled “5 cm |

## Continuous layers

- **Elevation:** cividis colour map (perceptually uniform, colour-blind safe),  $-0.5$  m to  $+2.5$  m relative to the ego ground plane; steps  $\geq 8$  cm get a thin white edge.
- **Traversability cost:** single-hue ramp from transparent (0, free) to #D55E00 vermillion (255, blocked); unknown cells keep the hatch.
- **Confidence:** grey-scale alpha mask over the semantic layer; low confidence looks washed out.

**UI colours (dark theme):** background #0E1116, panel #161B22, border #2A313C, primary text #E6EDF3, secondary text #8B949E, accent = sky blue #56B4E9 (shared with the fovea boundary). Status against targets: pass #3FB950, near limit #D29922, fail #F85149, always paired with an icon and the word.

## 5. Map rendering

The map draws each tier at its true cell size, so the viewer literally sees 5 cm cells give way to 50 cm cells at the boundary.

- **Default view:** top-down orthographic, vehicle fixed at centre, heading up, 60 m visible across by default; zoom from 20 m (cell grid visible) to 220 m (whole map).
- **3D view:** orbit camera; cells extruded to their ground or obstacle height, static obstacles as columns from ground to  $z_{\text{max}}$ , overhang clearance as a floating slab so low branches read at a glance.
- **Tier rendering:** each tier is one textured quad whose texture is the tier's layer array ( $400 \times 400$  texels), sampled with nearest-neighbour filtering so cell edges stay crisp. No per-cell geometry in 2D.
- **Cell grid lines:** shown only when a cell is  $\geq 6$  px on screen, in 6% white, so zoomed-in views show the lattice and zoomed-out views stay clean.
- **Fovea boundary:** dashed sky-blue square at  $\pm 10$  m (and  $\pm 25$  m in the graded profile) with a small tag “5 cm | 50 cm” on its top edge.
- **Range marks:** thin labelled ticks at 10, 25, 50 and 100 m along the heading axis; no full circles, which would clash with the square tiers.

- **Ego vehicle:** a to-scale footprint outline in white, with a short heading chevron.
- **Dynamic agents:** class colour, white outline, velocity arrow scaled to 1 s of travel; a faint trail of the last 5 positions (off under reduced motion).
- **Stale cells:** opacity falls with age after 1 s, reaching 40% at the 2 s staleness limit, so old information looks old.
- **Legend:** bottom-left of the map, lists only the layers currently on, with the class swatches from section 4 and the cell sizes of each tier.

## 6. Components

The right rail stacks five components from most to least glanceable: KPI tiles first, then the charts that explain them.

| Component            | Shows                                                             | Encoding                                                                                          | Updates                                                |
|----------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| KPI tiles (4)        | FPS, p95 latency, map memory, memory saving ×                     | Large tabular number, unit, target underneath, status icon + word                                 | Every frame, 4 Hz smoothing                            |
| Memory comparison    | Foveated vs uniform 2.5D vs voxel vs 2D occupancy                 | Dot plot on a log axis with values in a right column (same form as Architecture section 9)        | Every frame                                            |
| Latency by stage     | Preprocess, inference, projection, fusion, publish                | Stacked horizontal bar for the current frame + 30 s sparkline of total with the 50 ms target line | Every frame                                            |
| Accuracy by distance | mIoU per band: 0–10, 10–25, 25–50, 50–100 m                       | Vertical bars from zero, band labels, target line at 70%                                          | Per sequence (bench) or rolling 100 frames with labels |
| Integrity badge      | Points in vs points binned, cells updated                         | “All 128,406 points binned” in green, or the loss count in red                                    | Every frame                                            |
| Cell inspector       | Hovered cell's tier, size, class + confidence, heights, cost, age | Floating card near the cursor, monospace values                                                   | On hover, pin on click                                 |

**KPI tile anatomy:** label (12 px, secondary) → value (32 px, tabular numerals) + unit (14 px) → target line (“target ≤ 50 ms”) → status icon and word. Tiles never change size when numbers change.

## 7. Interaction and view modes

Every control has a keyboard shortcut so a presenter can drive the demo without hunting for buttons.

| Control                             | Action                                                                                         | Shortcut                    |
|-------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------|
| View: Top-down / 3D / Split compare | Split compare shows foveated and uniform grids side by side, cameras locked, memory under each | T / 3 / C                   |
| Layer toggles                       | Semantic, Elevation, Traversability, Dynamic, Confidence                                       | 1 – 5                       |
| Tier outline                        | Show or hide the fovea boundary and cell grid                                                  | G                           |
| Playback                            | Play / pause, step one frame, jump 10 frames                                                   | Space, ←, →, Shift + arrows |
| Timeline scrubber                   | Drag to any frame; markers show frames that failed a check                                     | mouse                       |
| Cell inspector                      | Hover to inspect, click to pin, Esc to clear                                                   | mouse, Esc                  |
| Follow / free camera                | Lock camera to vehicle or pan freely                                                           | F                           |
| Profile switch                      | Spec 2-tier / Graded 3-tier (re-projects the current frame)                                    | P                           |
| Snapshot                            | Save PNG + the frame's metrics JSON for the report                                             | S                           |

**Demo script the layout supports:** open on the top-down semantic view → zoom into a curb inside 10 m to show 5 cm detail → zoom out across the boundary → press C for split compare and read the memory saving → press 2 for elevation → point to FPS and latency tiles.

## 8. Typography, theming and accessibility

### Type

| Role                 | Font                             | Size / weight                 |
|----------------------|----------------------------------|-------------------------------|
| KPI value            | JetBrains Mono, tabular numerals | 32 px / 600                   |
| Panel title          | Inter                            | 14 px / 600                   |
| Body, labels         | Inter                            | 13 px / 400                   |
| Axis ticks, captions | Inter                            | 12 px / 400, secondary colour |
| Inspector values     | JetBrains Mono                   | 12 px / 400                   |

Numbers always use tabular figures so digits do not jitter as values update.

**Themes.** Dark is the default for demos and projectors. The light theme swaps UI tokens only; class colours keep their hue, with fills raised to 70% opacity on the light background.

## Accessibility

- Text contrast  $\geq 4.5:1$  against its panel in both themes; KPI values  $\geq 7:1$ .
- Class colours are Okabe-Ito, and no meaning rests on colour alone: moving = outline + arrow, unknown = hatch, status = icon + word.
- Full keyboard control (section 7) and visible focus rings in the accent colour.
- `prefers-reduced-motion` turns off trails, pulses and number tweening.
- Minimum 12 px text anywhere at 1080p; charts carry direct labels instead of relying on hover.

## 9. Motion, rendering performance and implementation

The browser renders at 60 fps while data arrives at 10–20 Hz; the two clocks are decoupled so the map moves smoothly without faking data.

- **Pose interpolation.** Between map frames, the ego pose is interpolated so the camera glides; cell contents change only when a new frame lands.
- **Delta updates.** The WebSocket sends each tier’s layer arrays as compressed binary (about 5 MB raw per frame for the 2-tier map, far less compressed), uploaded straight into GPU textures; no JSON per cell.
- **Instancing.** 2D tiers are textured quads; the 3D view uses instanced boxes only for occupied obstacle and agent cells.
- **Number motion.** KPI values tween over 150 ms, no bounce; charts append points, never re-animate whole series.
- **Back-pressure.** If the browser is more than one frame behind it skips to the newest frame and shows a small “display skipped N frames” note, separate from pipeline FPS.

- **Stack.** React + TypeScript shell; deck.gl (BitmapLayer per tier, custom layer for the boundary) for 2D and three.js for 3D; charts in D3 or visx; FastAPI WebSocket bridge (Architecture section 8).
- **Budget.** Render  $\leq 8$  ms per browser frame on an integrated GPU at 1080p; dashboard memory  $\leq 300$  MB.